

Yannik Schuler

Algebraic Geometer & Mathematical Physicist

Email: [yannik.schuler\[at\]math\[dot\]ethz\[dot\]ch](mailto:yannik.schuler[at]math[dot]ethz[dot]ch)
Web: yannikschuler.github.io

Employment

- Sep 2024 – Dec 2027 **Postdoc**
ETH Zürich, Department of Mathematics
Funded partly by a Walter Benjamin Fellowship of the DFG (Jan 2026–Dec 2027)
In Rahul Pandharipande’s research group
- Mar 2024 – Aug 2024 **Research Associate**
University of Cambridge, DPMMS
In Dhruv Ranganathan’s research group

Education

- Nov 2020 – Jun 2024 **PhD Mathematics**
University of Sheffield
Supervisor: Andrea Brini
Thesis: “Topics in Gromov–Witten theory”
- Apr 2017 – Jun 2020 **MSc Physics**
University of Bonn
Thesis: “Higher quantum Airy structures in two-dimensional topological gravity”
under supervision of Albrecht Klemm and Gaëtan Borot
- Apr 2017 – Mar 2019 **BSc Mathematics**
University of Bonn
Thesis: “Representations of modulated quivers”
under supervision of Jan Schröer
- Oct 2013 – Sep 2016 **BSc in Physics**
University of Bonn
Thesis: “Gravitational waves in view of the discovery at LIGO”
under supervision of Albrecht Klemm

Publications

Papers

- Michel van Garrel, Navid Nabijou, and Yannik Schuler. “Gromov–Witten theory of bicyclic pairs”. *Trans. Am. Math. Soc.* 379.5 (2026), pp. 3353–3402. DOI: 10.1090/tran/9581.
- Yannik Schuler. “The log-open correspondence for two-component Looijenga pairs”. *Forum Math. Sigma* 13 (2025). Id./No. e102, p. 30. DOI: 10.1017/fms.2025.10042.
- Andrea Brini and Yannik Schuler. “On quasi-tame Looijenga pairs”. *Commun. Num. Theor. Phys.* 17.2 (2023), pp. 313–341. DOI: 10.4310/CNTP.2023.v17.n2.a3.
- Gaëtan Borot, Reinier Kramer, and Yannik Schuler. “Higher Airy structures and topological recursion for singular spectral curves”. *Ann. Inst. Henri Poincaré Comb. Phys. Interact.* (2023). DOI: 10.4171/AIHPD/168.

Pre-Prints

- Daniel Holmes and Yannik Schuler. “Experiments with membranes, maps and sheaves”. 2026. arXiv: 2609 . 031 52 [math . AG].
- Alessandro Giacchetto, Rahul Pandharipande, and Yannik Schuler. “Membranes and Maps”. 2026. arXiv: 2609 . 02409 [math . AG].
- Yannik Schuler. “Gromov–Witten invariants and membrane indices of fivefolds via the topological vertex”. 2026. arXiv: 2603 . 24256 [math . AG].
- Andrea Brini and Yannik Schuler. “Refined Gromov–Witten invariants”. 2024. arXiv: 2410 . 00118 [math . AG].

Talks & Presentations

Conferences & workshops

Jul 2026	<i>Membranes and Maps</i> , Conference “Refinements in Enumerative Geometry and Physics” (ICTS Bengaluru, IN)
May 2026	<i>Fivefold perspective on the GW/DT correspondence</i> , Conference “Hilbert in Mirror” (SIMIS Shanghai, CN)
Jun 2025	<i>Curves in 5-folds and conjectures on Hodge integrals</i> , Conference “Helvetic Algebraic Geometry Seminar” (Les Diablerets, CH)
Jan 2025	<i>Membranes and Maps</i> , Conference “Algebra and Quantum Geometry of BPS quivers” (Les Diablerets, CH)
Sep 2024	<i>Maps, Sheaves and Membranes</i> , Conference “Duality, Geometry and Physics” (Bernoulli center, Lausanne, CH)
Sep 2024	<i>A worldsheet definition of the refined topological string</i> , Conference “Geometry and Physics of Mirror Symmetry” (University of Sheffield, UK)
May 2024	<i>Gromov–Witten theory and the refined topological string</i> , Conference “Real and Logarithmic Enumerative Geometry” (Oberwolfach, DE)
Jul 2022	<i>The Log-Local-Open Correspondence</i> , Program “K-theory, algebraic cycles and motivic homotopy theory” (INI Cambridge, UK)
Apr 2022	<i>Correspondences in enumerative geometry</i> , Student talk at “LMS Regional Meeting” (University of Birmingham, UK)

Seminars

May 2026	<i>Membranes and Maps</i> , Heidelberg Algebraic Geometry Seminar (University of Heidelberg, DE)
Nov 2025	<i>Equivariant topological strings on fivefolds</i> , Physical Mathematics Seminars (University of Geneva, CH)
Oct 2025	<i>Stable maps to fivefolds</i> , Harvard–MIT AG Seminar (MIT, US)
Oct 2025	<i>Refined Gromov–Witten invariants</i> , Geometry, Symmetry, and Physics Seminar (Yale University, US)
Oct 2025	<i>Membranes and Maps</i> , Informal Mathematical Physics Seminar (Columbia University, US)
Oct 2025	<i>Quintuple Hodge Integrals</i> , Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)
Aug 2025	<i>Refined Gromov–Witten invariants</i> , Moscow–Beijing topology seminar (online)
Oct 2024	<i>Refined mirror symmetry of local $\mathbb{P}^2$</i> , Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)
May 2024	<i>Log vs Open Gromov–Witten theory</i> , Fanosearch Seminar (Imperial College, UK)
May 2024	<i>Tangents to the nodal cubic</i> , Calf meeting (University of Bath, UK)
Apr 2024	<i>Tangents to the nodal cubic</i> , Junior Geometry Seminar (University of Cambridge, UK)
Apr 2024	<i>Gromov–Witten theory and the refined topological string</i> , Algebra Seminar (University of Edinburgh, UK)

Dec 2023	<i>Gromov–Witten theory and the refined topological string</i> , Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)
Nov 2023	<i>Gromov–Witten theory from the fivefold perspective</i> , Mathematical Physics Seminar (HU Berlin, DE)
Nov 2023	<i>Equivariant Strings and Gromov–Witten theory</i> , Algebraic Geometry Seminar (Utrecht University, NL)
Nov 2023	<i>Equivariant Curves and Strings</i> , MAGIC seminar (Imperial College, UK)
Oct 2023	<i>Tangents to the Nodal Cubic</i> , Geometry and Physics Seminar (University of Sheffield, UK)
May 2023	<i>Gromov–Witten theory from the fivefold perspective</i> , Geometry and Mathematical Physics Seminar (University of Birmingham, UK)
Feb 2023	<i>The Log-Local-Open Correspondence</i> , Algebraic Geometry Seminar (Cambridge University, UK)

Teaching Experience

Sep 2025 –	Lecturer ETH Zürich • Bachelor Seminar “Quiver Representations” (Fall 2025) • Master Seminar “Gromov–Witten theory” (Fall 2026)
Apr 2020 – Sep 2023	Postgraduate Teaching Assistant University of Sheffield • Demonstrator & Marker for “Core Foundation Mathematics” (Prep Course) • Demonstrator & Marker for “Mathematical modelling” (BSc) • Demonstrator & Marker for “Foundations of Pure Mathematics” (BSc) • Marker for “Advanced Calculus and Linear Algebra” (BSc)
Apr 2015 – Sep 2020	Teaching Assistant University of Bonn • Demonstrator & Marker for “Mathematics for physicists” (BSc) • Demonstrator & Marker for “Mathematics for computer scientists” (BSc) • Demonstrator & Marker for “Theoretical classical mechanics” (BSc) • Demonstrator & Marker for “Theoretical electrodynamics” (BSc) • Demonstrator & Marker for “Theoretical statistical physics” (BSc) • Head Tutor for “General Relativity” (MSc)

Student supervision

Bachelor theses

- J. Corti, Spring 2026, *The McKay Correspondence*
- A. Kalman, Fall 2025, *Burnside’s Character Formula for Hurwitz Numbers*

Master theses

- A. Dänzer, Fall 2025, *GW theory of a higher-dimensional generalization of the resolved conifold*

Academic service

Referee	Full reports for <i>Annales Henri Poincaré</i> , <i>Communications in Mathematical Physics</i> , <i>International Mathematics Research Notices</i> , <i>Journal of Algebraic Geometry</i> , <i>Transactions of the American Mathematical Society</i> Quick opinions for <i>Advances in Mathematics</i> , <i>Journal de l’École polytechnique Mathématiques</i>
---------	---

Reviewer for *zbMATH Open* and the AI mathematics benchmark *IMProofBench*

Scholarships

- Walter Benjamin Fellowship of the Deutsche Forschungsgemeinschaft (Jan 2026 – Dec 2027)
- Scholarship of the Bonn-Cologne Grad. School of Physics and Astronomy (Oct 2017 – Sep 2019)
- University of Sheffield Faculty of Science Scholarship (Nov 2020 – May 2024)